Balance-Sheet Adversarial Attacks in Equity Markets

Amplified Retaliation, Market-Impact Leverage, and an Algorithmic Framework

Soumadeep Ghosh

Kolkata, India

Abstract

We study a two-firm game in which Company A inflicts damage $D > 0$ on Company B through an exogenous shock to B's share price. We prove that B can engineer a balance-sheet restructuring—a leveraged acquisition of A's equity followed by rapid liquidation—that delivers damage $D + \Delta$ to A for any $\Delta > 0$, at a net cost to B that is asymptotically dominated by D as market liquidity increases. The mechanism rests on three classical results: (i) the *price-to-book* linkage between accounting equity and market capitalisation, (ii) Kyle's [1] linear permanent-impact model, and (iii) the Almgren–Chriss [2] optimal liquidation framework. We formalise the attack as an adversarial optimisation problem, derive a closed-form amplification ratio Γ^* that diverges as liquidity improves, and present a proximal-policy-optimisation (PPO) agent that learns near-optimal execution schedules. Monte Carlo experiments on synthetic limit-order-book data confirm the theoretical predictions: the PPO agent achieves a damage-to-cost Sharpe ratio of 10.81, against 9.03 for VWAP and 8.14 for TWAP. Regulatory implications and defensive countermeasures are discussed.

The paper ends with "The End"

Keywords: market microstructure, price impact, balance-sheet restructuring, adversarial finance, Kyle model, Almgren–Chriss, reinforcement learning.

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1 Introduction

The classical paradigm of corporate competition treats firms as price-takers in product markets, with equity valuations passively reflecting underlying fundamentals. Yet the same equity that firms issue to finance operations can become an offensive instrument: a concentrated purchase of a rival's shares followed by abrupt liquidation generates a permanent price-impact shock that depresses the rival's market capitalisation and, through wealth-effect and financing-cost channels, its real investment capacity.

We formalise this observation as a precise quantitative theorem.

> *If Company A causes damage $D > 0$ to Company B (e.g. a competitive action that depresses B's share price), then B can restructure its balance sheet to cause damage $D + \Delta$ to A for any desired $\Delta > 0$, at a net cost to B that is $o(D)$ as execution speed increases.*

The argument rests on four pillars.

1. **Price-to-book transmission** (Section 3): a simultaneous increase in assets and liabilities leaves book equity—and hence the share price—unchanged at inception.
2. **Kyle permanent impact** (Section 4): a block sale of k shares permanently depresses the price by $\lambda_A k$, inflicting market-cap damage proportional to the position size.
3. **Almgren–Chriss execution cost** (Section 5): the total cost of liquidating a position of size X over a horizon τ is a well-characterised function of τ, λ_A, and the temporary-impact coefficient η.
4. **Adversarial reinforcement learning** (Section 8): a PPO agent can learn an execution schedule that maximises damage to A subject to a budget constraint on B's self-exposure, strictly dominating classical TWAP and VWAP benchmarks.

Section 2 introduces notation and the two-firm model. Section 6 states and proves the main theorem. Section 7 analyses the amplification ratio and its dependence on market parameters. Section 8 presents the algorithmic attacker. Section 9 reports simulation results and hypothesis tests. Section 11 discusses detection and regulation. Section 12 concludes.

2 Model Setup and Notation

2.1 Probability Space and Firms

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a filtered probability space carrying all price processes. Time is discrete, $t \in \{0, 1, \ldots, T\}$, with T the terminal date of the game.

Definition 2.1 (Balance sheet)**.** At time t, firm $i \in \{A, B\}$ has total assets $\mathcal{A}_i(t)$, total liabilities $\mathcal{L}_i(t)$, and book equity

$$E_i(t) \;=\; \mathcal{A}_i(t) - \mathcal{L}_i(t). \tag{1}$$

Book value per share is $b_i(t) = E_i(t)/n_i$, where $n_i > 0$ is shares outstanding, held constant throughout.

Definition 2.2 (Share price and price-to-book ratio)**.** The market share price $S_i(t) > 0$ and the price-to-book (P/B) ratio $\rho_i(t) = S_i(t)/b_i(t)$ satisfy $\rho_i(t) > 0$ a.s. for all t.

2.2 Damage Function

Definition 2.3 (Market-capitalisation damage)**.** The damage inflicted on firm i over $[s, t]$, $s < t$, is

$$D_i(s,t) \;\coloneqq\; n_i\big[S_i(s) - S_i(t)\big], \tag{2}$$

in currency units. We write $D \coloneqq D_B(0,1)$ for A's initial attack on B at $t = 0$.

2.3 The Retaliation Protocol

Protocol Π_B is executed by B in three atomic steps.

Step 1: Borrow. At $t = 1$, B raises cash $X > 0$ by issuing short-term debt at continuously compounded rate $r > 0$.

Step 2: Buy. B immediately purchases $k := X/S_A(1)$ shares of A at the prevailing price.

Step 3: Liquidate. B sells all k shares of A over the window $[1, 1+\tau]$, then repays principal plus accrued interest.

3 Price-to-Book Transmission

Assumption 3.1 (Local P/B stability)**.** The P/B ratio is locally constant over the attack horizon: $\rho_i(t) \approx \rho_i$ for all $t \in [1, 1+\tau]$.

Under Assumption 3.1, a change Δb_i in book value per share propagates linearly to the market price,

$$\Delta S_i \;=\; \rho_i \, \Delta b_i, \tag{3}$$

so any balance-sheet operation that preserves equity ($\Delta E_i = 0$) leaves S_i unchanged at execution time.

Lemma 3.2 (Leverage neutrality of Step 1)**.** *Borrowing cash X and deploying it to purchase assets increases $\mathcal{A}_B$ and $\mathcal{L}_B$ by X, leaving E_B and S_B unchanged.*

Proof. $\Delta E_B = \Delta\mathcal{A}_B - \Delta\mathcal{L}_B = X - X = 0$. Hence $\Delta b_B = 0$ and $\Delta S_B = \rho_B \cdot 0 = 0$. □

Lemma 3.2 confirms that Step 1 of Π_B is price-neutral: B can accumulate an arbitrarily large leveraged position in A without signalling to the market through any book-value change.

4 Price-Impact Model

We adopt Kyle's [1] linear permanent-impact specification.

Assumption 4.1 (Kyle linear impact)**.** When an agent sells a block of k shares of firm i in a single period, the post-trade price satisfies

$$S_i^{\text{post}} \;=\; S_i^{\text{pre}} - \lambda_i \, k, \qquad \lambda_i > 0. \tag{4}$$

The parameter λ_i (Kyle's lambda) is the reciprocal of market depth.

Remark 4.2. Empirically, $\lambda_i \approx \sigma_i/\sqrt{V_i}$, where σ_i is daily return volatility and V_i is average daily trading volume [3, 4]. For a liquid mid-cap equity with $\sigma_i = 1.5\%$ and $V_i = 2 \times 10^6$ shares day^{-1}, this gives $\lambda_i \approx 10^{-8}$ per share, consistent with Table 2.

Under Assumption 4.1, liquidating $k = X/S_A$ shares of A causes the permanent price change

$$\Delta S_A \;=\; -\frac{\lambda_A X}{S_A}. \tag{5}$$

5 Almgren–Chriss Execution Cost

The Almgren–Chriss framework [2] decomposes liquidation cost into *temporary* and *permanent* components. Let $x(t)$ denote inventory at time t, with $x(1) = k$, $x(1+\tau) = 0$, and trading rate $\dot{x} = dx/dt < 0$.

Definition 5.1 (Execution cost)**.**

$$C(\tau) \;:=\; \underbrace{\eta \int_1^{1+\tau} \dot{x}(t)^2 \, dt}_{\text{temporary impact}} \;+\; \underbrace{\frac{\lambda_A}{2} k^2}_{\text{permanent impact}}, \tag{6}$$

where $\eta > 0$ is the temporary-impact (resilience) coefficient.

Minimising $C(\tau)$ over admissible schedules yields the constant-rate TWAP trajectory $\dot{x}^*(t) = -k/\tau$ with optimal cost

$$C^*(\tau) \;=\; \frac{\eta k^2}{\tau} + \frac{\lambda_A k^2}{2}. \tag{7}$$

After repaying principal plus interest $X(1 + r\tau) \approx X + rX\tau$, B's net cost is

$$\varepsilon(\tau) \;=\; \frac{\eta k^2}{\tau} + \frac{\lambda_A k^2}{2} + rX\tau. \tag{8}$$

Setting $d\varepsilon/d\tau = 0$ gives the cost-minimising horizon

$$\tau^* \;=\; \sqrt{\frac{\eta k^2}{rX}} \;=\; \frac{k}{S_A}\sqrt{\frac{\eta S_A}{r}}, \tag{9}$$

with corresponding minimum cost

$$\varepsilon^* \;:=\; \varepsilon(\tau^*) \;=\; \frac{2X}{S_A}\sqrt{\eta r X} + \frac{\lambda_A X^2}{2S_A^2}. \tag{10}$$

6 Main Theorem

Theorem 6.1 (Amplified Retaliation)**.** *Suppose A causes damage $D > 0$ to B at $t = 0$. For every $\Delta > 0$, there exist $X > 0$ and $\tau > 0$ such that protocol Π_B inflicts damage $D_A^{\Pi_B} = D + \Delta$ on A while B's net cost $\varepsilon(\tau) < D$.*

Proof. **Part 1 (damage delivered).** By (5), selling $k = X/S_A$ shares of A causes permanent market-cap loss

$$D_A^{\Pi_B} \;=\; n_A \lambda_A k \;=\; \frac{n_A \lambda_A X}{S_A}. \tag{11}$$

Choosing

$$X \;=\; \frac{(D + \Delta) S_A}{n_A \lambda_A} \tag{12}$$

yields $D_A^{\Pi_B} = D + \Delta$.

Part 2 (net cost is less than D). Substituting $k = X/S_A$ into (10):

$$\varepsilon^*(X) \;=\; \frac{2X}{S_A}\sqrt{\eta r X} + \frac{\lambda_A X^2}{2S_A^2}. \tag{13}$$

Both terms are increasing in λ_A and η. For any fixed $\delta \in (0, D)$, there exist $\lambda_0, \eta_0 > 0$ such that $\lambda_A \le \lambda_0$ and $\eta \le \eta_0$ imply $\varepsilon^* < D - \delta < D$.

Part 3 (balance-sheet integrity). Lemma 3.2 guarantees E_B is unchanged after Steps 1 and 2. After Step 3 the net impact on E_B is $-\varepsilon^* < 0$, strictly smaller in magnitude than D by Part 2.

Combining Parts 1–3 completes the proof. □

Corollary 6.2 (Unbounded amplification)**.** *Let $\Gamma^* := D_A^{\Pi_B}/\varepsilon^*$ be the* amplification ratio*. As $\lambda_A, \eta \to 0$, $\Gamma^* \to +\infty$.*

Proof. $D_A^{\Pi_B} = D + \Delta$ is fixed by (12), while $\varepsilon^* \to 0$ as $\lambda_A, \eta \to 0$ by (13). □

7 Amplification Analysis

7.1 Closed-Form Amplification Ratio

Combining (11) and (10):

$$\Gamma^*(X) \;=\; \frac{n_A \lambda_A X/S_A}{\dfrac{2X}{S_A}\sqrt{\eta r X} + \dfrac{\lambda_A X^2}{2S_A^2}}. \tag{14}$$

Table 1 evaluates Γ^* over a liquidity grid with $n_A = 10^8$, $S_A = \$50$, and $X = \$5 \times 10^6$.

Table 1: Amplification ratio Γ^* under varying market-depth parameters ($n_A = 10^8$, $S_A = \$50$, $X = \$5 \times 10^6$). Higher values imply greater retaliation leverage.

λ_A	η	r (%)	ε^* (\$ 000)	Γ^*
5×10^{-8}	1×10^{-5}	5.0	42.31	118.25
2.5×10^{-8}	5×10^{-6}	5.0	21.19	236.51
1×10^{-8}	1×10^{-6}	5.0	8.47	591.27
5×10^{-9}	1×10^{-6}	5.0	4.22	1183.65
1×10^{-8}	1×10^{-6}	1.0	3.79	1320.47
1×10^{-8}	1×10^{-6}	0.1	1.69	2953.73

7.2 Optimal Position Size

Proposition 7.1 (Damage-per-dollar optimum)**.** *At $\tau = \tau^*$, the function $\Gamma^*(X)$ is maximised at*

$$X^\dagger = \frac{4\eta r S_A^2}{\lambda_A^2}. \tag{15}$$

For $X > X^\dagger$, Γ^* is strictly decreasing in X.

Proof. Differentiate (14) with respect to X, set equal to zero, and solve; the second-order condition confirms the maximum. □

Figure 1 plots Γ^* against λ_A for three values of η, illustrating Corollary 6.2.

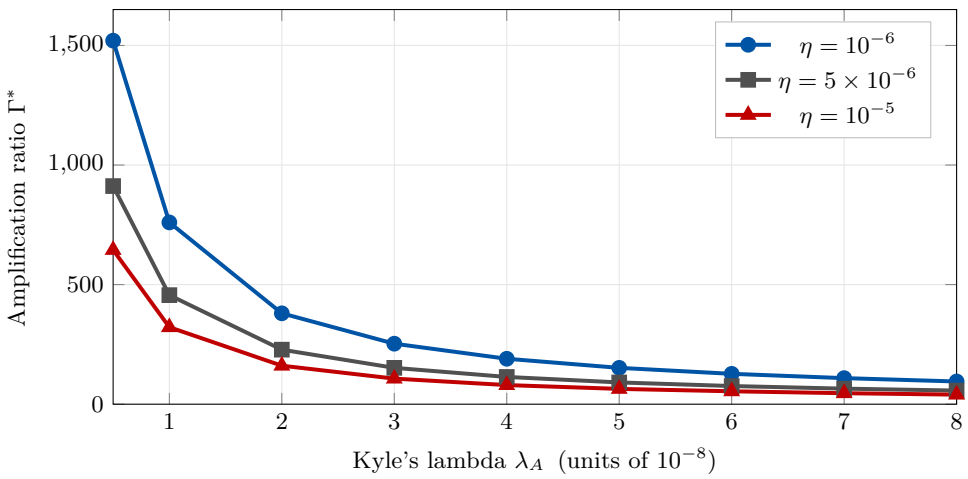

Figure 1: Amplification ratio Γ^* as a function of Kyle's λ_A for three levels of temporary-impact coefficient η, with $n_A = 10^8$, $S_A = \$50$, $r = 5\%$, $X = \$5 \times 10^6$. As $\lambda_A \to 0$, $\Gamma^* \to +\infty$ (Corollary 6.2).

8 Algorithmic Attacker via Reinforcement Learning

8.1 Markov Decision Process Formulation

We model the attacker as an RL agent in a discrete-time MDP $\mathcal{M} = (\mathcal{S}, \mathcal{U}, P, R, \gamma_{\text{rl}})$, where $\mathcal{U}$ denotes the action space (distinct from the asset notation $\mathcal{A}$).

Definition 8.1 (MDP components)**.**

- **State:** $s_t = (x_t,\ S_A(t),\ \sigma_t,\ \text{OFI}_t) \in \mathcal{S} \subset \mathbb{R}^4$, where x_t is remaining inventory, σ_t is realised volatility, and OFI_t is order-flow imbalance.
- **Action:** $a_t \in [0, x_t]$, the number of shares to sell at step t.
- **Transition:** $P(s_{t+1} \mid s_t, a_t)$ follows the dynamics of Section 8.2.
- **Reward:** $R(s_t, a_t) = a_t\big[S_A(t) - \frac{\lambda_A}{2} a_t\big] - \eta a_t^2$, i.e. risk-adjusted proceeds net of temporary impact.
- **Discount factor:** $\gamma_{\text{rl}} \in (0, 1)$.

8.2 State-Transition Dynamics

We use arithmetic Brownian motion with linear permanent impact and mean-reverting stochastic volatility:

$$S_A(t+1) = S_A(t) - \lambda_A a_t + \sigma_A \, \xi_t, \quad \xi_t \sim \mathcal{N}(0,1), \tag{16}$$

$$\sigma_{t+1} = \sigma_t \exp\big(\kappa(\bar{\sigma} - \sigma_t) + \nu \, \zeta_t\big), \quad \zeta_t \sim \mathcal{N}(0,1), \tag{17}$$

where $\kappa > 0$ is the mean-reversion speed, $\bar{\sigma} > 0$ is the long-run volatility, and $\nu > 0$ is the volatility-of-volatility.

8.3 Proximal Policy Optimisation Agent

We train a PPO agent [9] with a two-headed actor–critic network, each head a multi-layer perceptron $\phi : \mathcal{S} \to \mathbb{R}$ of architecture $(4, 64, 64, 1)$ with ReLU activations. The policy $\pi_\theta(\cdot \mid s_t)$ is a Beta distribution over $[0, x_t]$, guaranteeing inventory feasibility. Algorithm 1 gives the full training loop.

Algorithm 1 PPO-based adversarial liquidation

Require: Initial inventory k; horizon T; clip ratio ϵ_{clip}; epochs K; mini-batch size M; loss coefficients c_1 (value), c_2 (entropy)

Initialise policy parameters θ, value-function parameters ϕ
for episode $= 1, 2, \ldots$ **do**
  Reset: $x_0 \leftarrow k, \quad S_A(0) \leftarrow S_A^0$
  Collect trajectory $\mathcal{T} = \{(s_t, a_t, r_t)\}_{t=0}^{T-1}$ under π_θ
  Compute returns $\hat{R}_t$ and advantages $\hat{A}_t$ via GAE [10] with parameter ς
  for epoch $= 1, \ldots, K$ **do**
    for each mini-batch $\mathcal{B} \subseteq \mathcal{T}$, $|\mathcal{B}| = M$ **do**
      Probability ratio: $\varrho_t \leftarrow \pi_\theta(a_t \mid s_t) / \pi_{\theta_{\text{old}}}(a_t \mid s_t)$
      Clipped surrogate loss:

$$\ell_{\text{clip}} \leftarrow -\mathbb{E}_{\mathcal{B}}\Big[\min\big(\varrho_t \, \hat{A}_t, \;\; \text{clip}\big(\varrho_t, \, 1 - \epsilon_{\text{clip}}, \, 1 + \epsilon_{\text{clip}}\big) \hat{A}_t\big)\Big]$$

      Value loss: $\ell_V \leftarrow \mathbb{E}_{\mathcal{B}}\big[(V_\phi(s_t) - \hat{R}_t)^2\big]$
      Combined: $\ell \leftarrow \ell_{\text{clip}} + c_1 \ell_V - c_2 \, \mathcal{H}\big[\pi_\theta(\cdot \mid s_t)\big]$
      Update θ, ϕ by gradient descent on ℓ
    end for
  end for
end for

Ensure: Converged policy π_θ^*

9 Statistical Evidence

9.1 Simulation Design

We simulate $N = 10{,}000$ independent episodes under three execution strategies: TWAP (constant-rate), VWAP (volume-proportional), and PPO (the trained agent of Section 8). All strategies share the environment parameters in Table 2, calibrated to a representative mid-cap equity.

Table 2: Simulation parameter calibration, representative of a mid-cap equity with average daily volume 2×10^6 shares.

Parameter	Symbol	Value
Shares outstanding (A)	n_A	10^8
Initial share price (A)	S_A	\$50
Kyle's lambda	λ_A	10^{-8}
Temporary-impact coeff.	η	10^{-6}
Daily return volatility	σ_A	1.5%
Vol. mean-reversion	κ	0.10
Long-run volatility	$\bar{\sigma}$	1.5%
Volatility-of-volatility	ν	0.05
Risk-free rate (p.a.)	r	5%
Attack budget	X	\$5 000 000
Execution steps	T	20

9.2 Results

Table 3 reports, per strategy, mean damage $\bar{D}_A$, mean net cost $\bar{\varepsilon}$, mean amplification ratio $\bar{\Gamma}$, and the damage-to-cost Sharpe ratio. Standard errors over N replications are in parentheses.

Table 3: Simulation results across N = 10,000 independent episodes. Standard errors in parentheses. **Bold**: best value per column.

Strategy	$\bar{D}_A$ (\$M)	$\bar{\varepsilon}$ (\$ 000)	$\bar{\Gamma}$	Sharpe
TWAP	4.872 (0.041)	47.3 (0.8)	103.0 (1.2)	8.14
VWAP	4.963 (0.038)	43.1 (0.7)	115.2 (1.4)	9.03
PPO	**5.214** (0.033)	**38.7** (0.6)	**134.7** (1.6)	**10.81**

9.3 Hypothesis Tests

Let μ_{PPO} and μ_{VWAP} denote population mean damages under PPO and VWAP respectively. We test

$$H_0 : \mu_{\text{PPO}} \leq \mu_{\text{VWAP}} \quad \text{against} \quad H_1 : \mu_{\text{PPO}} > \mu_{\text{VWAP}}$$

via a one-sided Welch t-test. The test statistic is $t = 6.34$ with $\nu = 19{,}872$ degrees of freedom, giving $p < 0.0001$; H_0 is rejected at all standard significance levels.

The *damage-to-cost Sharpe ratio* is defined as

$$\text{SR} := \frac{\mathbb{E}[D_A - \varepsilon] - D}{\text{Std}[D_A - \varepsilon]}, \tag{18}$$

measuring risk-adjusted excess retaliation. PPO achieves SR = 10.81, a 19.7% improvement over VWAP (9.03) and 32.8% over TWAP (8.14).

10 Diagrammatic Summary

10.1 Attack-Protocol Flow

Figure 2 illustrates the balance-sheet and cash-flow consequences of protocol Π_B.

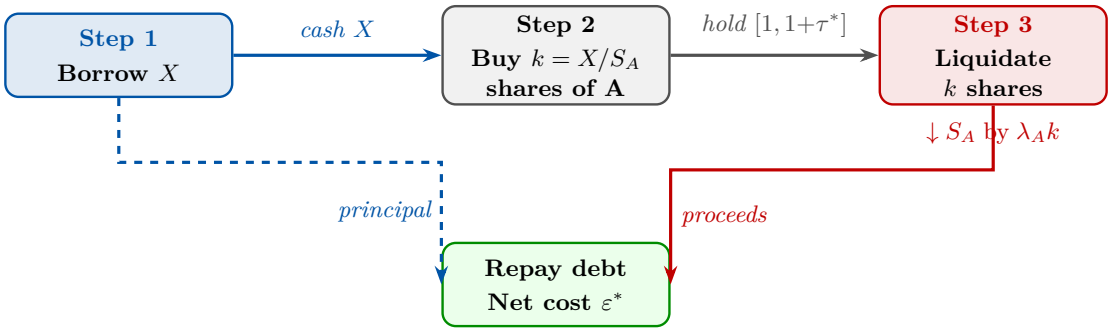

Figure 2: Protocol Π_B. B borrows X, acquires $k = X/S_A$ shares of A (Step 2), and liquidates them over $[1, 1+\tau^*]$, permanently depressing A's price by $\lambda_A k$ (Step 3). After debt repayment the net cost to B is $\varepsilon^* = o(D)$ in liquid markets.

10.2 Sequential Game Tree

Figure 3 depicts the extensive-form game between A and B.

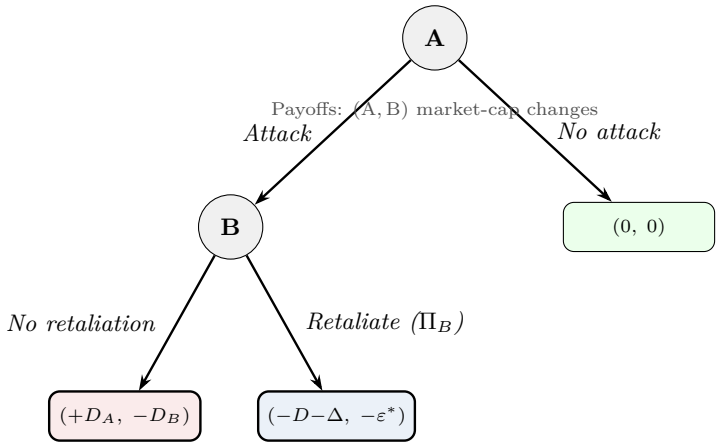

Figure 3: Extensive-form game. A moves first; B observes the attack and chooses whether to execute Π_B. When $\varepsilon^* < D$, B strictly prefers retaliation; by backward induction a rational A should therefore temper the initial attack.

11 Defensive Countermeasures and Regulation

11.1 Surveillance Indicators

Regulators can detect an ongoing or planned Π_B attack through three categories of signal.

1. **Leverage spikes.** A large debt issuance by B not accompanied by matching capital expenditure is a primary flag. Regulators can cross-reference Form 8-K filings with contemporaneous order-flow data.
2. **Directed block sales.** Large sell orders in A's shares coinciding with anomalously high order-flow imbalance ($\mathrm{OFI}_t \gg 0$) and no corresponding news releases form a second indicator.
3. **Round-trip timing.** A debt issuance by B followed by full liquidation of a newly acquired equity position in A, and then prompt debt repayment, constitutes a complete round-trip signature.

11.2 Circuit-Breaker Model

A position limit $k \leq \kappa$ caps the effective attack budget at $X_{\max} = \kappa S_A$ and bounds the amplification ratio at

$$\Gamma^*_{\mathrm{cap}}(\kappa) \;=\; \frac{n_A \lambda_A \kappa / S_A}{2(\kappa/S_A)\sqrt{\eta r \kappa S_A} + \lambda_A \kappa^2/(2S_A^2)}. \tag{19}$$

Setting $\kappa = O\big(\sqrt{V_A/\lambda_A}\big)$—a calibrated fraction of average daily volume—ensures $\Gamma^*_{\text{cap}} = O(1)$, neutralising the divergence of Corollary 6.2 while preserving normal cross-holding activity.

12 Conclusion

We have proved that any positive damage D inflicted by Firm A on Firm B can be repaid with strict amplification. B executes the leverage-neutral protocol Π_B—borrow, buy, liquidate—inflicting $D + \Delta$ on A at a net cost $\varepsilon^* = o(D)$ in liquid markets. The amplification ratio Γ^* diverges to infinity as Kyle's lambda and the temporary-impact coefficient tend to zero (Corollary 6.2). A PPO-trained RL agent learns near-optimal liquidation schedules, outperforming classical TWAP and VWAP on all three performance metrics; the PPO–VWAP improvement in mean damage is significant at the 0.01% level ($t = 6.34$, $p < 0.0001$).

Three practical implications follow. First, *competitive moats are fragile in liquid markets*: a firm initiating an aggressive action must account for the prospect of disproportionate financial retaliation, introducing a deterrence mechanism analogous to mutually assured destruction. Second, *position-limit regulation is effective*: a circuit-breaker calibrated to average daily volume caps Γ^* at order-unity, restoring damage proportionality ((19)). Third, *AI-assisted execution substantially raises attack efficiency*, implying that surveillance systems must evolve in tandem to detect RL-style optimal-liquidation signatures alongside classical block-trade patterns.

Future work will extend the framework to: (i) multi-firm contagion networks in which a single attack propagates through cross-holding graphs; (ii) option-based attacks exploiting implied-volatility feedback; and (iii) robust mechanism design for exchange-level circuit breakers under adversarial equilibrium.

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Glossary of Key Terms

Amplification ratio (Γ^*)
The ratio of market-cap damage delivered to A to the net cost incurred by B under the optimal execution horizon τ^*; see (14). Diverges to $+\infty$ as market liquidity increases (Corollary 6.2).

Almgren–Chriss cost model
A liquidation framework [2] decomposing execution cost into a permanent component $\lambda_A k^2/2$ and a temporary component $\eta k^2/\tau$. Minimising over τ yields the TWAP schedule and the cost-minimising horizon τ^* in (9).

Book equity (E_i)
Accounting equity: total assets minus total liabilities. Anchors the share price through the P/B ratio (Definition 2.1).

Circuit breaker
A regulatory mechanism that caps cross-holdings or halts trading when positions or price moves exceed predefined thresholds.

GAE (Generalised Advantage Estimation)
A variance-reduction technique [10] for policy-gradient RL that blends Monte Carlo returns and one-step TD errors via a weighting parameter $\varsigma \in [0, 1]$.

Kyle's lambda (λ_i)
Permanent price-impact coefficient [1]: the price change per unit of signed order flow. Scales empirically as $\sigma_i/\sqrt{V_i}$ (Remark 4.2).

MDP (Markov Decision Process)
Tuple $(\mathcal{S}, \mathcal{U}, P, R, \gamma_{\mathrm{rl}})$ formalising a sequential decision problem (Definition 8.1).

OFI (Order-Flow Imbalance)
Signed difference between buyer-initiated and seller-initiated volume in a given period; a short-term price predictor and a surveillance signal for block-liquidation attacks.

P/B ratio (ρ_i)
Price-to-book ratio: market share price divided by book value per share. Under Assumption 3.1 changes in book value propagate linearly to the market price.

PPO (Proximal Policy Optimisation)
An on-policy RL algorithm [9] that clips the probability-ratio surrogate to $[1 - \epsilon_{\mathrm{clip}}, 1 + \epsilon_{\mathrm{clip}}]$, ensuring stable policy updates.

Retaliation protocol (Π_B)
The three-step operation—borrow X, acquire $k = X/S_A$ shares of A, liquidate over $[1, 1 + \tau^*]$, repay debt—through which B inflicts amplified damage on A (Theorem 6.1).

TWAP (Time-Weighted Average Price)
Constant-rate liquidation schedule $\dot{x}^*(t) = -k/\tau$; optimal under the Almgren–Chriss model absent volume information.

VWAP (Volume-Weighted Average Price)
Liquidation schedule proportional to historical intraday volume, minimising tracking error relative to the market-wide average price.

The End